Sign Misattribution in Step-Independent Agent RL: Dual Mechanism Diagnosis and Episode-Level Fix

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Abstract

Multi-turn agent reinforcement learning requires credit assignment across long interaction trajectories, yet current frameworks default to step-independent windowing that decomposes trajectories into independently optimized windows. This introduces a structural credit assignment pathology driven by two mechanisms: *policy information loss*, which truncates context and imposes a constant performance ceiling, and *advantage sign misattribution*, which causes gradient updates to reflect episode outcomes rather than local action quality, producing cumulative degradation. At p=12.5% with group size K=8, theory predicts 80.9% of gradient updates for locally correct actions carry incorrect sign under binary reward; empirical measurement under penalty reward yields a 52% credit agreement rate, indistinguishable from random chance. A 2×2 factorial separates the mechanisms: eliminating sign misattribution converts catastrophic collapse into bounded learning; removing information loss enables generalization. Episode-level Group Relative Policy Optimization (GRPO), treating the full trajectory as a single optimization sample, resolves both mechanisms. A systematic 11-seed evaluation reveals that stable training is necessary but not sufficient: 73% of seeds avoid collapse, but only 36% achieve statistically significant out-of-distribution (OOD) generalization (p<0.01; 24–31% OOD vs. 6.7% supervised fine-tuning (SFT) baseline). Preliminary single-seed cross-scale verification at 8B shows both failure and fix (90.3% OOD vs. collapse to 1.5%), and cross-environment validation on ScienceWorld provides evidence the collapse extends beyond ALFWorld.

1 Introduction

Reinforcement learning from sparse terminal reward has emerged as a promising paradigm for training LLM agents on multi-turn interactive tasks spanning 30–50 interaction steps (Kim et al., 2025; Bai et al., 2024; Wang et al., 2022). Group Relative Policy Optimization (GRPO) (Shao et al., 2024), with its critic-free design, has become the algorithm of choice. Yet a gap persists between the promise of RL-driven agent improvement and the reality of training outcomes: recent agent RL systems (Wang et al., 2025, 2026a; Qin et al., 2026; Feng et al., 2025) share a common, unscrutinized implementation pattern—decomposing trajectories into fixed-size windows, each optimized independently—and frequently report training instability without diagnosing its root cause.

We show this *step-independent* default introduces a fundamental credit assignment pathology. Step-independent training partitions trajectories into non-overlapping h-step windows, each optimized via GRPO. This produces a two-phase collapse: task success drops to 0% while action validity initially remains high, then continued training destroys even basic formatting. Standard remedies—dense reward shaping, discount tuning—fail. Concurrent methods that improve per-step credit quality (GiGPO (Feng et al., 2025), GAGPO (Zhu et al., 2026), Turn-PPO (Li et al., 2026a)) address symptoms but not the structural root cause: the windowing itself.

We trace the collapse to two mechanisms at different layers. ***Policy information loss*** (M1) operates at the inference layer: each window lacks access to earlier observations, imposing a constant performance ceiling. ***Advantage sign misattribution*** (M2) operates at the training layer: under binary reward, the advantage sign reflects episode success rather than local action quality. At p=12.5% with K=8, Proposition 2 predicts 80.9% of gradient updates for locally correct windows carry incorrect sign under binary reward (the empirically measured credit agreement rate drops to 52% under penalty reward; Section 4.4)—sign errors cause active divergence, not merely slow convergence. Our primary contribution is diagnostic:

we identify, formalize, and empirically validate a previously uncharacterized pathology in the dominant training paradigm. Episode-level GRPO—not a new method, but the original GRPO design—eliminates both mechanisms by treating the full trajectory as a single optimization sample: 17.2% mean OOD success where step-independent collapses to 0%, and 90.3% OOD at 8B versus 1.5% collapse.

Our contributions:

1. **Dual mechanism diagnosis.** We formalize *information loss* and *sign misattribution* as distinct failure modes with different temporal behaviors (constant ceiling versus cumulative degradation) and derive the probability of incorrect gradient direction as a function of baseline success rate and group size.
2. **Empirical separation.** A 2×2 factorial isolates the two mechanisms, a window-size sweep reveals a credit assignment cliff, direct sign measurement confirms the theoretical prediction (52% credit agreement ≈ random), and preliminary cross-scale (8B) and cross-environment (ScienceWorld) evidence supports generality.
3. **Episode-level fix: necessary but not sufficient.** GRPO's original episode-level design resolves both mechanisms, but a systematic 11-seed evaluation reveals that 73% of seeds avoid collapse while only 36% achieve statistically significant OOD generalization ($p<0.01$). We analyze this seed-dependent transfer gap and why the community defaulted to the broken step-independent paradigm.

2 Method

2.1 Problem Formulation and GRPO

We model the multi-turn agent as a partially observable Markov decision process (POMDP) where the policy $\pi_\phi(a_t \mid h_t)$ generates actions conditioned on interaction history $h_t = (o_1, a_1, \ldots, o_t)$. Episodes of T steps terminate with binary reward $R \in \{0, r_+\}$. GRPO (Shao et al., 2024) samples K rollouts per task and computes group-relative advantages:

$$\hat{A}_k = \frac{R_k - \mu}{\sigma + \epsilon}, \qquad \mu = \frac{1}{K}\sum_{j=1}^{K} R_j, \quad \sigma = \sqrt{\frac{1}{K}\sum_{j=1}^{K}(R_j - \mu)^2} \tag{1}$$

GRPO is episode-level by design: each rollout receives one return and one advantage, with no inherent temporal discounting.

2.2 Step-Independent Windowing

Agent RL frameworks must reconcile GRPO's episode-level design with the memory constraints of long trajectories. RAGEN (Wang et al., 2025) introduces **step-independent training**: the trajectory is partitioned into non-overlapping h-step windows, each receiving a discounted return:

$$R_k = \gamma^{(N-k)} \cdot R_{\text{episode}} \tag{2}$$

where $N = \lceil T/h \rceil$ is the number of windows. Each window is optimized as an independent GRPO sample. This reduces sequence length from $O(T)$ to $O(h)$ and multiplies the effective batch size, but structurally alters credit assignment.

2.3 Dual Mechanism Analysis

Two mechanisms jointly cause training failure under step-independent optimization.

2.3.1 Mechanism 1: Policy Information Loss

Step-independent training truncates context at both inference and training time. The policy $\pi_\phi(a_t \mid h_{\text{window}_k})$ conditions on only the local h-step history, losing access to earlier observations. With h=2, a window at step 8 has no access to the outcome of an exploration action at step 4, so the policy cannot learn strategies spanning more than h steps. This imposes a *constant* performance ceiling: no amount of optimization can recover absent information.

2.3.2 Mechanism 2: Advantage Sign Misattribution

Group normalization under step-independent training produces advantages whose sign reflects the *episode* outcome, not local action quality:

$$\text{sign}(\hat{A}_k) \approx \text{sign}(R_{\text{episode}} - \mu) \tag{3}$$

In failed episodes ($R_{\text{episode}} = 0$), all windows receive negative advantages when $\mu > 0$, including windows with locally correct actions. Conversely, successful episodes reinforce all windows, including those with suboptimal actions. Unlike M1's constant ceiling, sign misattribution is *cumulative*: each wrong-sign gradient step further erodes policy competence. The discount factor $\gamma^{(N-k)}$ introduces magnitude decay across positions, but sign

errors are strictly more harmful—magnitude decay slows convergence, while sign errors cause active divergence.

2.4 Episode-Level Training

Episode-level training resolves both mechanisms by treating the complete trajectory as a single GRPO sample. The full history h_t is available at every step (eliminating M1), and a single advantage $\hat{A}$ applies uniformly to all tokens (eliminating M2). Uniform reinforcement of successful trajectories preserves good sub-strategies; window-level sign errors actively destroy them. This is GRPO's original design; step-independent windowing is the deviation.

2.5 Factorial Experiment Design

To separate the two mechanisms, we implement a 2×2 factorial varying observation scope and advantage granularity: **Cell A** (truncated h=2 + step-level advantage; both mechanisms active), **Cell B** (truncated h=2 + episode-level advantage; only M1 active), and **Cell D** (full trajectory + episode-level advantage; neither active). The fourth cell—**Cell C** (full trajectory + step-level advantage; only M2 active)—would isolate sign misattribution without information loss. We attempted this configuration but encountered persistent training instability (Section 4.2; Appendix C). Comparing A vs. B isolates sign misattribution; comparing B vs. D isolates information loss.

3 Theoretical Analysis

Sign misattribution is the more fundamental failure mechanism because it causes active divergence, not merely slow convergence. We derive the probability of incorrect gradient direction below; full proofs appear in Appendix A.

Definition 1 (Locally correct action). *An action a_t is **locally correct** if it would receive positive advantage under an oracle that observes the full episode history h_t and has access to the true action-value function. Equivalently, window k contains locally correct actions if replacing them (holding all other windows fixed) would decrease the probability of episode success.*

Proposition 1. *Under step-independent training with binary episode reward $R_{episode} \in \{0, r_+\}$, the probability that window k's advantage $\hat{A}_k$ has the correct sign (aligned with local action quality) approaches the base success rate p as the number of rollouts K per group increases.*

Proof sketch. In a failed episode ($R_{\text{episode}} = 0$), all windows receive $R_k = 0$. If any rollout in the group succeeded, $\mu > 0$, so every window in the failed episode receives $\hat{A}_k < 0$—including windows with locally correct actions. The probability of receiving the wrong sign in a failed episode is $1 - (1-p)^{K-1}$ (at least one group member succeeded). As K grows, this approaches $1 - p$, and the overall probability of correct sign approaches p. □

The key distinction is temporal. Information loss (M1) imposes a constant ceiling determined by h; sign misattribution (M2) is cumulative—each wrong-sign gradient step erodes policy competence below that ceiling. A wrong-sign gradient moves the optimizer *away* from good policies; a reduced-magnitude gradient from discounting ($\gamma^{(N-k)}$) still points toward good policies. Sign errors produce active divergence; magnitude errors merely slow convergence (Appendix A.1).

Proposition 2. *Let p denote the episode success rate and K the number of rollouts per group. Under step-independent training with binary episode reward, the fraction of gradient updates with correct sign is:*

$$P(\text{correct} \mid \text{update}) = \frac{p}{1 - (1-p)^K} \qquad (4)$$

Derivation. Three mutually exclusive cases determine gradient direction for a window with locally good actions: (1) episode succeeded (probability p): correct sign; (2) episode failed but group contains a success (probability $(1-p)[1-(1-p)^{K-1}]$): wrong sign; (3) entire group failed (probability $(1-p)^K$): zero gradient. Conditioning on non-zero gradient (Cases 1 and 2) yields Equation 4.

At p=12.5%, K=8, only 19.1% of gradient updates have the correct sign (Figure 1). In the large-K limit, $P(\text{correct} \mid \text{update}) \to p$: ∼87.5% of updates for locally correct windows point in the wrong direction.

Severity scales with task difficulty. Sign misattribution becomes *more* severe as the task gets harder (Equation 4): at p=50%, $P(\text{correct} \mid \text{update}) = 50.2\%$—barely above random; at p=5%, only 5.1% of updates are correct. Multi-turn agent tasks with low baseline success rates are precisely where credit assignment matters most, yet where step-independent training produces the worst gradient signal. Increasing K does not help:

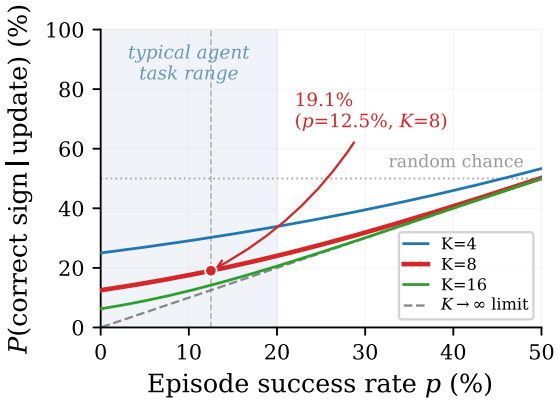


Figure 1: Probability of correct gradient sign as a function of episode success rate p (Proposition 2). At typical agent task success rates ($p < 0.2$, shaded), the majority of gradient updates carry incorrect sign regardless of group size K. At ALFWorld's baseline (p=12.5%, K=8), only 19.1% of updates are correct. The dashed diagonal shows the large-K limit $P_{\text{correct}} \to p$.

more updates occur (fewer all-fail groups), but each remains equally likely to carry the wrong sign.

Reward saturation. When all K rollouts receive the same binary reward, $\sigma = 0$ and $\hat{A}_k = 0$, producing zero gradient. At p=12.5%, K=8, the all-fail probability is $(1-p)^K \approx 34.4\%$; combined with the negligible all-success case ($p^K < 10^{-7}$), roughly one-third of batches yield no gradient at all. This compounds sign misattribution: of the $\sim$66% of batches that produce a gradient, only 19.1% have the correct sign. The effective learning rate is $0.66 \times 0.19 \approx 12.6\%$ of nominal—well below the regime where convergence guarantees hold.

Cumulative divergence. Sign misattribution produces a systematic gradient bias. Let θ^* denote an optimal policy parameterization. At each update, the expected gradient under step-independent training is:

$$\mathbb{E}[\nabla_\theta \mathcal{L}] = P_c \cdot g_+ + (1 - P_c) \cdot g_- \qquad (5)$$

where $P_c = P(\text{correct} \mid \text{update})$, g_+ points toward θ^*, and g_- points away. When $P_c < 0.5$ (which holds for $p < 0.5$; Equation 4), the expected gradient points *away* from good policies. Under the simplifying assumption that correct and incorrect gradient components have comparable magnitude, after t steps: $\|\theta_t - \theta^*\| \geq \|\theta_0 - \theta^*\| + t \cdot \eta(1 - 2P_c)\|g\|$, where η is the learning rate. This is not slow convergence but monotonic divergence—each step increases the distance to good policies, explaining the two-phase collapse observed empirically.

4 Experiments

Step-independent training collapses while episode-level training succeeds; a factorial design separates the two mechanisms; and window size h modulates severity.

4.1 Setup

We evaluate on ALFWorld (Shridhar et al., 2021), a text-based household environment with six task types requiring sequential reasoning over up to 50 steps. Reward is binary: $R \in \{0, r_+\}$. The base policy is Qwen3-1.7B fine-tuned on 2,546 expert demonstrations (SFT baseline: 12.5% in-distribution, 0% pre-SFT). At p=0.125 with K=8, Proposition 2 predicts $\sim$80.9% of gradient updates for locally correct windows carry incorrect sign. All conditions use RAGEN (Wang et al., 2025) (v0.1, accessed May 2026)[1] with GRPO (Shao et al., 2024) (K=8, lr= 10^{-6}, β=0.01, γ=0.95, 15 steps; full hyperparameters in Appendix B). We report success rate on a 134-task OOD set (primary) and 32-task in-distribution set (secondary). Seed counts are noted per condition.

4.2 Factorial Mechanism Separation

We implement the 2 × 2 factorial (Section 2.5): **Cell A** (both mechanisms active; h=2, step-level advantage), **Cell B** (M1 only; h=2, episode-level advantage), and **Cell D** (neither; full trajectory, episode-level advantage). Two seeds each. The fourth cell—**Cell C** (full observation, step-level advantage)—would isolate M2 without M1; we predict partial improvement over Cell A, since removing information loss enables gradient alignment but step-level advantage may still introduce noise. We attempted this configuration but encountered persistent training instability (Appendix C).

4.3 Results

Cell A collapses in two phases (Table 1): task success drops to 0% within 3–9 steps while action validity initially remains above 80%, then continued sign misattribution destroys formatting (validity: 89.3% → 43.1% between steps 2–4, reaching 0.06% by step 15).Cell B eliminates sign misattribution while retaining information loss: training

[1]Code available at https://github.com/RAGEN-AI/RAGEN.

Table 1: 1.7B results on ALFWorld (best checkpoint per seed, evaluated on held-out test sets). In-Dist: 32-task set; OOD: 134-task set. Cell A reports best seed; Cell B reports best checkpoint (step 5). Cell D reports mean of two seeds (In-Dist range: 28.1–40.6; OOD range: 10.4–23.9).

Condition	Mechanisms	In-Dist (%)	OOD (%)
SFT baseline	—	12.50	6.72
Cell A (h=2, step)	M1+M2	0	0
Cell B (h=2, episode)	M1 only	18.75	5.97
Cell D (full, episode)	neither	**34.38**	**17.16**
Dense reward	M1+M2	0	0
REINFORCE++	neither	12.50	—

success reaches 55% (seed 2), but OOD generalization is bounded by the information-loss ceiling (best OOD: 5.97% at step 5). Both seeds converge to the same OOD ceiling (0.75% at step 15) despite different training trajectories (20% vs. 55%), confirming the bottleneck is architectural rather than statistical.

Cell D reaches 17.2% mean OOD success (range: 10.4–23.9%) with monotonic improvement across both seeds, the only condition that substantially exceeds the SFT baseline.

The A→B comparison isolates M2: eliminating sign misattribution converts catastrophic collapse into bounded learning. The B→D comparison isolates M1: removing the information-loss ceiling enables OOD generalization (+11.2pp mean). Both mechanisms are necessary to explain the full failure pattern. Figure 2(a) visualizes these trajectories.

Additional conditions. Dense reward, REINFORCE++, and discount-factor sweep experiments all confirm that sign misattribution—not reward sparsity or discounting—drives the failure (Appendix C).

Window-size sweep. A credit assignment cliff between h=10 and h=20 separates collapse from recovery (Figure 2b; seed 1): h=2: 0%, h=5: 0%, h=10: 6.25%, h=20: 15.6%, episode-level: 23.88%. This cliff pattern is observed in a single seed; multi-seed verification is needed to confirm its robustness. Only episode-level training achieves substantial generalization.

4.4 Sign Misattribution Verification

We operationalize local correctness (Definition 1) through episode outcome: in a successful episode, all windows with valid actions are labeled locally correct. The credit agreement rate—the fraction of such windows where GRPO's advantage sign is positive—is 52.07% under step-independent training (h=2), indistinguishable from 50% random chance. Advantage sign correlates near-perfectly with episode outcome (r>0.99) but shows near-zero dependence on local quality (point biserial r=0.068). Combined with γ=1.0 collapse (Appendix C), these measurements confirm sign misattribution as the causal driver.

Table 2: Episode-level GRPO across 11 seeds (134 OOD tasks, τ=0.4, best checkpoint per seed). Category: C=Collapsed, S=Non-collapsed (stagnant), G=Generalized (Fisher exact p<0.01 vs. SFT). Wilson 95% CI. SFT baseline: 6.72% (9/134). Seed2 achieves p=0.02, falling just outside the conservative p<0.01 threshold, and is classified as Stagnant. Step-5 results for all seeds in Appendix E.

Seed	Step	Cat.	OOD (%)	95% CI	vs SFT
seed2048	5	C	8.21	[4.6, 14.2]	NS
seed42	5	C	12.69	[8.0, 19.4]	NS
seed8	5	C	13.43	[8.6, 20.3]	NS
seed200	5	S	10.45	[6.3, 16.8]	NS
seed300	5	S	10.45	[6.3, 16.8]	NS
seed100	5	S	11.19	[6.9, 17.6]	NS
seed100	10	S	12.69	[8.0, 19.4]	NS
seed2	15	S	15.67	[10.5, 22.8]	p=0.02
seed1	5	G†	23.88	[17.4, 31.8]	p<0.001
seed5	10	G	25.37	[18.8, 33.3]	p<0.001
seed6	15	G	29.10	[22.2, 37.1]	p<0.001
seed4	15	G	30.60	[23.5, 38.7]	p<0.001

† RAGEN built-in eval; checkpoint unavailable for re-evaluation.

4.5 Episode-Level Training: Seed Variability

Episode-level GRPO resolves the dual-mechanism failure, but training outcomes remain seed-dependent. We systematically evaluate N=11 random seeds under identical hyperparameters, revealing a 2×2 structure: *training stability* (collapsed vs. non-collapsed) × *OOD generalization* (stagnant vs. generalized).

We define **collapsed** seeds as those where in-distribution success declines >50% from peak and remains below baseline for ≥3 consecutive steps. **Non-collapsed** seeds are the complement. Among non-collapsed seeds, we further distinguish **generalized** (Fisher exact p<0.01 vs. SFT on the 134-task OOD set) from **stagnant** (not significant vs. SFT). Table 2 reports results for each seed at its best-performing checkpoint.

Three distinct outcomes emerge. 3/11 seeds collapse (27%), with response length inflating to

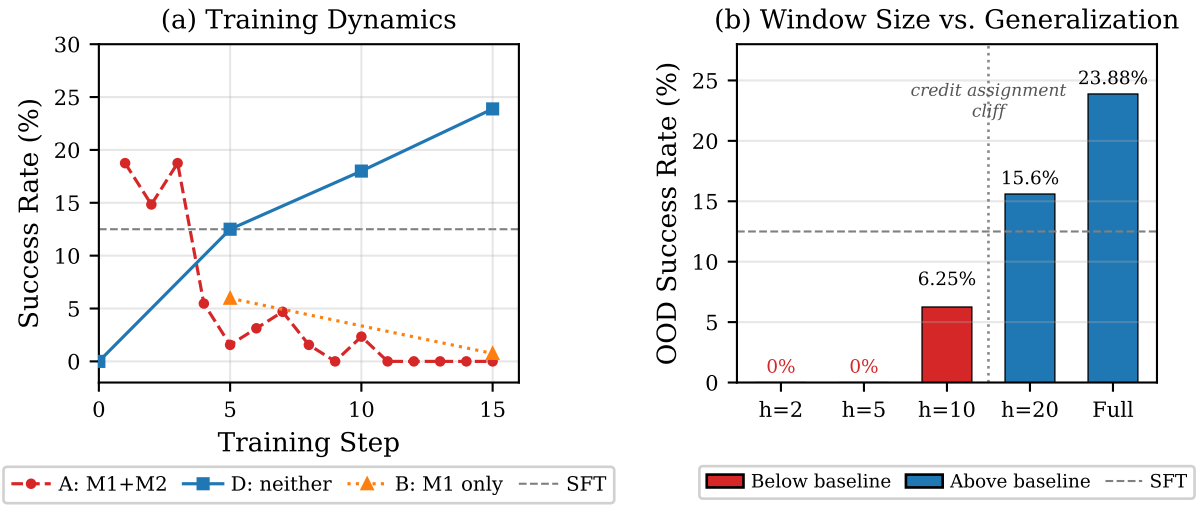


Figure 2: (a) Training dynamics across factorial conditions (metric type indicated in legend). Cell A collapses within 5 steps; Cell B shows bounded learning; Cell D improves monotonically. (b) Window size h vs. OOD success (seed 1; cf. Table 1 for two-seed mean). A credit assignment cliff between h=10 and h=20 separates collapse from recovery. Hatched bars indicate collapsed conditions.

14,000–16,000 tokens (Appendix D); 4/11 are non-collapsed but stagnant (36%), with OOD indistinguishable from SFT (10.45–15.67%, all NS at α=0.01); and 4/11 generalize (36%), reaching 23.88–30.60% OOD (p<0.001). No collapsed seed generalizes, confirming stable training is necessary. One seed (seed2) shows delayed learning: 15.67% OOD after 15 steps (p=0.02, borderline).

The stagnant–generalized gap is not explained by insufficient training: seed100 improves only 1.50pp from step 5 to step 10 (11.19% → 12.69%, both NS), while seed5 reaches 25.37% at step 10. Stable training is necessary but not sufficient for OOD generalization—the mechanism driving seed-dependent transfer remains an open question (Section 6).

4.6 Cross-Scale Verification

A single-seed experiment on Qwen3-8B provides preliminary evidence that both failure and fix persist at scale (Table 3). Cell A collapses within two steps (28.1% → 1.5%). Cell D reaches 96.9% in-distribution and 90.3% OOD at step 10, exceeding the SFT baseline (80.6% OOD) by 9.7pp. Restricting to h=2 drops in-distribution success from 68.75% to 31.25% (37.5pp gap). Single-seed due to Fully Sharded Data Parallel (FSDP) memory constraints; multi-seed verification is needed to confirm robustness.

In this single-seed experiment, the episode-level fix appears more effective at 8B (90.3% OOD vs. 17.2% mean at 1.7B; Figure 3a), suggesting larger models may have more capacity unlocked by correct credit assignment.

Table 3: 8B results on ALFWorld. SFT and h=2 use decoding temperature τ=0; RL uses τ=0.4. Cell A: step 2 (post-collapse). Cell D: step 10.

Method	In-Dist (%)	OOD (%)
SFT baseline	68.75	80.60
SFT h=2	31.25	—
Cell A: step-indep. (h=2)	28.1 → 1.5	—
Cell D: episode-level	**96.88**	**90.30**

4.7 Cross-Environment Validation

Step-independent training collapses on ScienceWorld (Wang et al., 2022) (30 task types, SFT baseline 30.1%): Cell A (h=2) drops to 0% task success throughout all 15 steps, despite the higher baseline (p=30.1% vs. ALFWorld's 12.5%).Restricting context to h=2 drops SFT success from 34.8% to 4.5% (30.3pp M1 gap), paralleling the 37.5pp gap at 8B ALFWorld (Figure 3b).These single-seed results provide preliminary evidence that both mechanisms operate beyond ALFWorld; multi-seed and multi-environment verification would strengthen this claim.

5 Related Work

RL for multi-turn LLM agents. Policy gradient methods—PPO (Schulman et al., 2017), GRPO (Shao et al., 2024), DPO (Rafailov et al.,

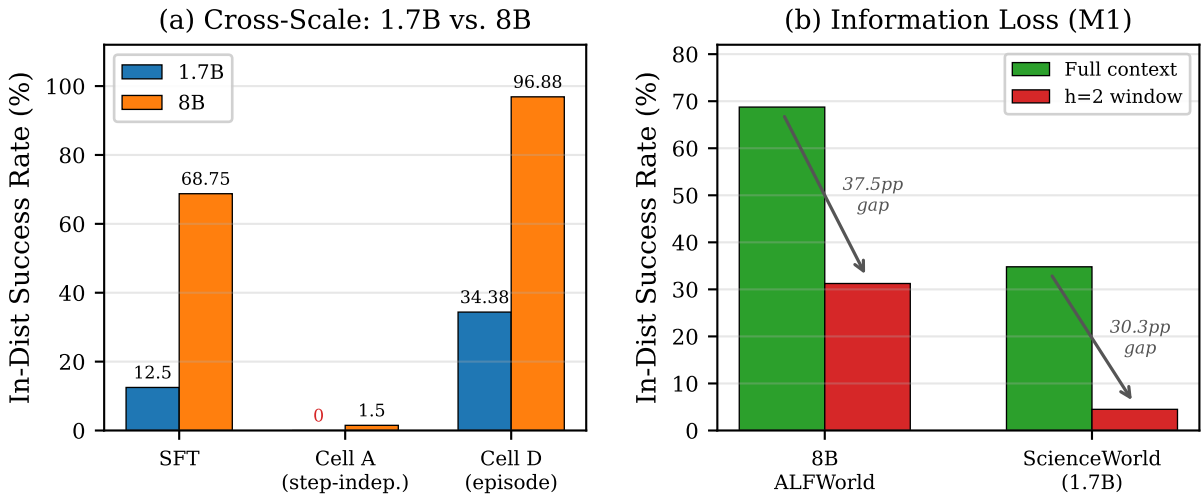


Figure 3: (a) Cross-scale verification (single seed): both collapse (Cell A) and fix (Cell D) are observed from 1.7B to 8B. The episode-level fix appears more effective at 8B (96.9% vs. 34.4% in-distribution). (b) Information loss (M1) quantification (single seed each): restricting context to h=2 drops SFT performance by 37.5pp at 8B ALFWorld and 30.3pp on ScienceWorld.

2023)—underpin RLHF (Ouyang et al., 2022) and have been extended to multi-turn agents (Yao et al., 2023) through step-independent windowing (Wang et al., 2025). Subsequent work improves algorithms (GiGPO (Feng et al., 2025)), reward design (Self-Guide (Wang et al., 2026a)), or training pipelines (SPEAR (Qin et al., 2026), DigiRL (Bai et al., 2024)), but all share a common assumption: trajectory decomposition into fixed windows is benign. We show this assumption is false—the windowing introduces a structural credit assignment pathology that no algorithmic improvement within the paradigm can fix.

Credit assignment in reinforcement learning. TD(λ) (Sutton, 1988), GAE (Schulman et al., 2016), and TRPO (Schulman et al., 2015) interpolate between bootstrapping and Monte Carlo returns; REINFORCE (Williams, 1992) uses full episode returns. Reward shaping (Ng et al., 1999) and hindsight experience replay (Andrychowicz et al., 2017) address sparse reward in classical RL; counterfactual baselines (Foerster et al., 2018) handle team reward credit assignment. Process reward models (Lightman et al., 2024; Wang et al., 2024) provide step-level supervision for reasoning tasks. In the low-success-rate regime ($p < 0.2$) typical of agent tasks, the classical bias-variance tradeoff breaks down: Proposition 2 shows sign errors under windowing grow as $1-p$, dominating any variance reduction that windowing provides.

Credit assignment and sign misattribution evidence. Several independent studies corroborate our diagnosis. IRC (Modecrua et al., 2026) reports per-turn advantages frequently disagree in sign with outcome advantages. MC-GRPO (Kim, 2025) measures a 5% sign-flip rate. GRPO-λ (Parthasarathi et al., 2025) proves value estimation error grows with position. OAR (Li et al., 2026b) identifies "reward aliasing" from coarse credit. Our contribution is to formalize this as a structural property of the training paradigm and to show it interacts with information loss in a vicious cycle.

Multi-turn credit assignment methods. Turn-PPO (Li et al., 2026a) introduces per-turn value estimation, GAGPO (Zhu et al., 2026) uses generalized advantage with a turn-level critic, MT-GRPO (Wei et al., 2025) applies Monte Carlo returns at the turn level, and SWEET-RL (Zhou et al., 2025) combines step-wise rewards with episode-level training. These methods improve per-step signal quality. Our work is complementary: we diagnose *why* step-independent training fails, informing when finer-grained credit is safe to use (only after establishing correct gradient direction via episode-level baselines).

GRPO collapse phenomena. RAGEN-2 (Wang et al., 2026b) proves advantage collapse when reward variance vanishes. GRPO-Collapse (Deng et al., 2025) identifies a death spiral in Search-R1. Gradient Coupling (Liu et al., 2025) argues the

failure is high input similarity across steps. Our dual-mechanism diagnosis unifies these reports: reward variance collapse is a special case of reward saturation (Section 3), and the death spiral follows from the M1–M2 vicious cycle.

6 Discussion

Generality and mechanism interaction. Sign misattribution arises whenever three structural properties co-occur: (a) step-level windowing, (b) sparse binary reward, and (c) group-relative normalization. Under these conditions, the advantage sign collapses to a proxy for episode outcome regardless of local quality, and severity scales with task difficulty ($P(\text{correct} \mid \text{update}) \to p$). M1 and M2 interact in a vicious cycle: M1 lowers p by preventing multi-step strategies, increasing wrong-sign updates under M2, which further degrades p—explaining why step-independent training produces active collapse rather than slow convergence.

Why step-independent became the default. Six contributing factors, from context length constraints to framework design inertia, are detailed in Appendix A.3.

Seed-dependent OOD generalization. The multi-seed evaluation (Section 4.5) reveals a bimodal distribution: a stagnant cluster (10–16% OOD, NS vs. SFT) and a generalized cluster (25–31%, $p<0.001$). Training duration does not explain the gap (seed100: $+1.50$pp from step 5 to 10; seed5: 25.37% at step 10), pointing to early training dynamics or initialization sensitivity. Identifying this mechanism is an important open question.

Practitioner guidance. Use episode-level credit when the full episode fits in context; otherwise use the largest feasible h ($h \geq 20$; $h \leq 10$ collapses). Finer-grained methods (Turn-PPO, GAGPO) should be applied only after establishing an episode-level baseline, as they assume correct gradient direction.

7 Conclusion

Step-independent training in multi-turn agent RL fails through two mechanisms: policy information loss imposes a performance ceiling, while advantage sign misattribution causes cumulative degradation: at $p=12.5\%$ with $K=8$, Proposition 2 predicts $\sim$80.9% incorrect-sign updates under binary reward (52% measured under penalty reward, consistent with near-random credit assignment). A 2×2 factorial separates the mechanisms; episode-level GRPO resolves both (17.2% mean OOD at 1.7B; preliminary single-seed 8B evidence shows 90.3% OOD). A systematic 11-seed evaluation reveals that stable training is necessary but not sufficient: 73% avoid collapse, but only 36% generalize ($p<0.01$). The central insight—sign errors are more destructive than magnitude imprecision—applies to any agent RL method combining step-level windowing, sparse binary reward, and group-relative normalization.

Limitations

Our experiments span two environments (ALFWorld, ScienceWorld) and two scales (1.7B, 8B); additional environments and scales would strengthen generality. The 8B and ScienceWorld results report single-seed runs due to computational constraints. Among 11 seeds at 1.7B, only 36% achieve statistically significant OOD generalization; we cannot rule out that stagnant seeds would eventually generalize with substantially more training steps, though seed100's minimal improvement from step 5 to step 10 suggests the bottleneck is not simply training duration. The mechanism driving seed-dependent transfer remains an open question. For very long episodes ($T > 100$) or tasks with many independent sub-goals, uniform episode-level advantage may limit efficiency even though it preserves direction correctness.

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A Theoretical Analysis: Full Derivations

A.1 Why Sign Misattribution is the Core Mechanism

The two mechanisms identified in Section 2.3 operate at different layers and produce qualitatively different failures. Policy information loss (Mechanism 1) operates at the inference layer, restricting the observable context and establishing a structural performance ceiling that is constant across training. Sign misattribution (Mechanism 2) operates at the training layer, corrupting gradient directions and progressively degrading performance below this ceiling.

Proof of Proposition 1. Define "correct sign" as follows: for a window containing locally beneficial actions, the correct sign of $\hat{A}_k$ is positive (reinforcement); for a window containing locally detrimental actions, the correct sign is negative (suppression). Under binary reward, the return for window k is $R_k = \gamma^{(N-k)} \cdot R_{\text{episode}}$ (Equation 2). In a failed episode ($R_{\text{episode}} = 0$), all windows receive $R_k = 0$ regardless of their local action quality. If the group contains at least one successful rollout, the group mean $\mu > 0$, and all windows in the failed episode receive negative advantages after normalization via Equation 1. This includes windows where the agent took locally correct actions. Now consider a window with good local actions. In a successful episode (probability p), the window receives positive $\hat{A}_k$, which is the correct sign. In a failed episode (probability $1 - p$), the window receives negative $\hat{A}_k$ whenever $\mu > 0$, which is the wrong sign. The probability that $\mu = 0$ (all K rollouts in the group failed) is $(1-p)^K$, which shrinks exponentially with K. As K grows, nearly every failed episode produces a non-zero group mean, and the window receives the wrong sign with probability approaching $1-p$. The probability of correct sign therefore approaches p. □

The distinction between sign error and magnitude error further clarifies why Mechanism 2 is catastrophic while Mechanism 1 is merely suboptimal. The discount factor $\gamma^{(N-k)}$ in Equation 2 introduces magnitude decay across window positions: for an episode with $T = 50$ steps and $h = 2$, the first window ($k = 1$) retains $\gamma^{N-1} = 0.95^{24} \approx 0.29$ of the terminal window's return. This attenuation reduces the gradient magnitude for early windows but preserves the gradient direction. A gradient with correct sign and reduced magnitude produces slow convergence, a condition recoverable through longer training. A gradient with wrong sign produces active divergence: the optimization pushes the policy away from good behaviors and toward bad ones.

A.2 Formal Probability Analysis

Derivation of Proposition 2. For a window with locally good actions, the gradient direction depends on $\text{sign}(\hat{A}_k)$, which depends on whether the episode succeeded and whether the group contains any successful rollouts. Three mutually exclusive cases arise.

Case 1: Episode succeeded (probability p). The window receives $R_k = \gamma^{(N-k)} \cdot r_+ > 0$. With at

least one failed rollout in the group, $R_k > \mu$ after normalization, producing $\hat{A}_k > 0$.[2] The gradient reinforces the window's actions. For a window with good local actions, this is the correct direction.

Case 2: Episode failed, group contains at least one success (probability $(1-p)[1-(1-p)^{K-1}]$). The window receives $R_k = 0 < \mu$, producing $\hat{A}_k < 0$. The gradient suppresses the window's actions. For a window with good local actions, this is the wrong direction.

Case 3: Episode failed, entire group failed (probability $(1-p)\cdot(1-p)^{K-1} = (1-p)^K$). All returns are zero, so $\mu = 0$ and $\sigma = 0$. The advantage is zero, producing no gradient update.

Summing Cases 1 and 3 yields $P(\text{correct sign or no update}) = p + (1-p)^K$. Since only Cases 1 and 2 produce non-zero gradients, the conditional probability follows as Equation 4.

Remark. Proposition 2 assumes binary episode rewards for analytical tractability. Under non-binary rewards (e.g., with per-step penalties for invalid actions), the group mean μ is nonzero even when all episodes fail, but the qualitative conclusion is preserved: advantage sign is determined by relative episode performance within the group, not by local action quality within individual windows. Section 4.4 validates this empirically using the full reward structure.

Penalty reward effect on credit agreement. Proposition 2 predicts 19.1% correct-sign rate (i.e., 80.9% incorrect) under pure binary reward. Our experiments include a -0.1 penalty per invalid action, making the reward non-binary. This penalty introduces variance within failed episodes: windows with valid actions receive $R_k = 0$ while windows with invalid actions receive $R_k = -0.1h$. Within a failed episode, the penalty-induced variance can occasionally produce positive $\hat{A}_k$ for windows with valid actions even when $\mu > 0$, since the normalization accounts for intra-group variance. This shifts the empirical credit agreement from the theoretical 19.1% toward 52%—still indistinguishable from the 50% random baseline and far below the level needed for convergence. The gap between 19.1% (binary theory) and 52% (penalty-reward measurement) thus reflects the reward structure difference, not a failure of the theoretical framework.

We evaluate these expressions at the empirical baseline of our experiments. At $p = 12.5\%$ and $K = 8$: $(1-p)^K = 0.875^8 \approx 0.344$, giving $P(\text{correct or no update}) \approx 0.469$. Of the 65.6% of cases that produce a gradient update, only $P(\text{correct} \mid \text{update}) = 0.125/0.656 \approx 19.1\%$ have the correct sign. As K increases to 16, $(1-p)^K \approx 0.118$, and $P(\text{correct} \mid \text{update}) \approx 14.2\%$. In the large-$K$ limit, $P(\text{correct} \mid \text{update}) \to p$, so approximately $1 - p = 87.5\%$ of all gradient updates for locally good windows point in the wrong direction.

A.3 Why the Community Defaults to Step-Independent Training

If step-independent training is structurally flawed for multi-turn agents, why has it become the default? We identify six contributing factors.

1. Context length constraints. Smaller language models (1B to 4B parameters) typically support context lengths of 2048 to 4096 tokens. Multi-turn episodes spanning 50 or more interaction steps easily exceed these limits. Step-independent windowing is a pragmatic response to a hardware constraint, not a deliberate algorithmic choice.

2. Token-level training inertia. Language model training is organized around predicting the next token over fixed-length sequences. When multi-turn RL inherits this paradigm, the natural unit of optimization becomes a fixed-length window rather than a variable-length episode. The mismatch between token-level optimization and episode-level credit assignment goes unquestioned because it aligns with existing training infrastructure.

3. Inherited framework default. The RAGEN framework (Wang et al., 2025) implements step-independent mode as its default configuration, not as one option among alternatives. Users inherit this choice without an explicit decision point, and default configurations propagate silently through downstream projects.

4. Memory and compute constraints. Episode-level training requires processing $O(T)$ tokens per sequence, incurring $O(T^2)$ attention cost with full self-attention. Step-independent training reduces this to $O(h)$ tokens and $O(h^2)$ attention cost per window. The computational savings are immediate and measurable, while the credit assignment failure is subtle and deferred until training collapse occurs.

5. Reward shaping as apparent fix. Dense reward shaping (per-step bonuses for intermediate goals) appears to address sparse reward problems.

[2] The edge case where all K rollouts succeed (probability p^K) is negligible at the baseline success rates considered ($p^8 < 10^{-7}$ at $p = 12.5\%$).

Our dense reward experiment (Section C) reveals a different failure mode: the agent maximizes format-level rewards (action validity rises from 60% to 97.5%) while task success remains at 0%. Reward shaping does not resolve sign misattribution; it creates new optimization targets that the agent can game without solving the underlying task.

6. Absence of negative results. When step-independent training fails, practitioners attribute the failure to insufficient hyperparameter tuning or the inherent difficulty of sparse rewards. Without published analyses that isolate the credit assignment mechanism as the root cause, the community cannot distinguish between "this problem is hard" and "this training paradigm is broken for this class of problems."

A.4 Connection to Classical Credit Assignment

Our findings connect to a long line of work on temporal credit assignment in reinforcement learning.

Temporal difference learning and eligibility traces. Sutton's TD(λ) (Sutton, 1988) provides a spectrum of credit assignment strategies parameterized by $\lambda \in [0, 1]$. At $\lambda = 0$, credit is assigned via one-step bootstrapped returns. At $\lambda = 1$, TD(λ) reduces to Monte Carlo returns. Step-independent training falls below this spectrum: windows are disconnected from each other, so temporal propagation of credit does not occur.

Generalized Advantage Estimation. GAE(λ, γ) (Schulman et al., 2016) extends TD(λ) with a separate discount factor. GAE(1, 1) produces undiscounted Monte Carlo returns and is equivalent to episode-level GRPO without a critic. The key difference is that GAE uses value function subtraction as the baseline, while GRPO uses group normalization.

REINFORCE. Williams' REINFORCE algorithm (Williams, 1992) computes policy gradients from complete episode returns with a learned baseline $b(s)$. Episode-level GRPO is equivalent to REINFORCE where the group mean μ replaces $b(s)$ and the group standard deviation σ provides normalization. The algorithmic structure is not new. Our contribution is the diagnosis: the community's standard adaptation of this algorithm for multi-turn LLM agents (step-independent windowing) silently breaks the credit assignment structure that REINFORCE assumes.

Multi-agent credit assignment analogy. Step-independent windows can be understood through cooperative multi-agent games. Each window acts as an independent "agent" sharing a team reward but optimized without knowledge of other windows' actions. The sign misattribution problem parallels team reward credit assignment in cooperative multi-agent RL: when the team fails, all agents receive identical penalties regardless of individual contribution. In multi-agent RL, this is a recognized problem with known solutions such as counterfactual baselines and value decomposition methods.

B Hyperparameters

Table 4: Shared RL training hyperparameters across all conditions.

Parameter	Value
Group size K	8
Learning rate	1×10^{-6}
KL penalty β	0.01
Discount γ	0.95
Invalid action penalty	0.1
Tasks per training step	16
Training steps	15

All 1.7B experiments were conducted on a single NVIDIA A100 80 GB GPU. Each training run (15 steps, K=8 rollouts per task, 16 tasks per step) required approximately 6–8 GPU-hours. The 11-seed evaluation totaled ∼80 GPU-hours. The 8B experiment used 4× A100 80 GB with FSDP, requiring ∼24 GPU-hours per run. ScienceWorld experiments used the same single-GPU setup as the 1.7B ALFWorld runs.

C Additional Conditions

All conditions use single seeds unless noted.

Dense reward. Step-independent h=2 with +0.15 per valid action. Action validity rises from 60% to 97.5% while task success remains 0%—the agent games format-level rewards without solving tasks (Figure 4b).

REINFORCE++. Episode-level with per-token running-mean baseline (no critic). REINFORCE++ matches the SFT baseline (12.50%) without improvement, suggesting that per-token advantage normalization alone does not help when the per-token baseline lacks a learned value function. We note that 15 training steps may be insufficient

for REINFORCE++ to converge; however, the Cell A collapse pattern represents active degradation rather than mere undertraining, as performance drops below the SFT initialization.

Discount factor sweep. $\gamma \in \{0.50, 0.80, 0.95, 1.00\}$ at h=5. All four collapse to 0% within 5 steps, including γ=1.0 where no magnitude decay exists. This confirms sign misattribution, not discounting, is the operative mechanism (Figure 4a).

Cell C attempt. Cell C (full observation + step-level advantage) would isolate M2 without M1. We attempted this configuration across six training runs with varying hyperparameters (KL penalty $\beta \in \{0.001, 0.01, 0.04\}$, learning rate $\in \{5\times10^{-7}, 10^{-6}\}$). All runs terminated due to training instability: KL divergence spiked above 10.0 within 2–3 steps, triggering NaN losses. We attribute this to the combination of full-length sequences (>4096 tokens) with step-level advantage normalization, which produces extreme advantage magnitudes at early window positions. Cell C remains an important gap in the factorial; addressing it likely requires architectural changes (e.g., gradient clipping per window position) beyond the scope of this diagnostic study.

D Response Length During Training

E Unified Step-5 Evaluation

To guard against cherry-picking of checkpoints, Table 5 reports all 11 seeds at a fixed evaluation point (training step 5). OOD results are included only for seeds with checkpoints evaluated under the D134 protocol (Section 4.1); remaining checkpoints were unavailable for post-hoc OOD evaluation.

Table 5: Episode-level GRPO: all 11 seeds at training step 5 (134 OOD tasks, 32 ID tasks, τ=0.4). SFT baseline: 12.5% ID, 6.72% OOD. OOD results marked "—" indicate checkpoints unavailable for post-hoc evaluation on the 134-task OOD set. ID results marked "—" indicate step-5 validation metrics not recorded in training logs.

Seed	Cat.	ID (%)	OOD (%)	vs SFT
seed2048	C	—	8.21	NS
seed8	C	28.13	13.43	NS
seed42	C	9.38	12.69	NS
seed200	S	—	10.45	NS
seed300	S	25.00	10.45	NS
seed100	S	31.25	11.19	NS
seed2	S	—	—	—
seed1	G†	—	23.88	p<0.001
seed4	G	—	—	—
seed5	G	—	—	—
seed6	G	21.88‡	—	—

† RAGEN built-in eval; checkpoint unavailable for re-evaluation.
‡ Registry value 21.9% (7/32); rounded to 21.88% for consistency with 32-task denominator.

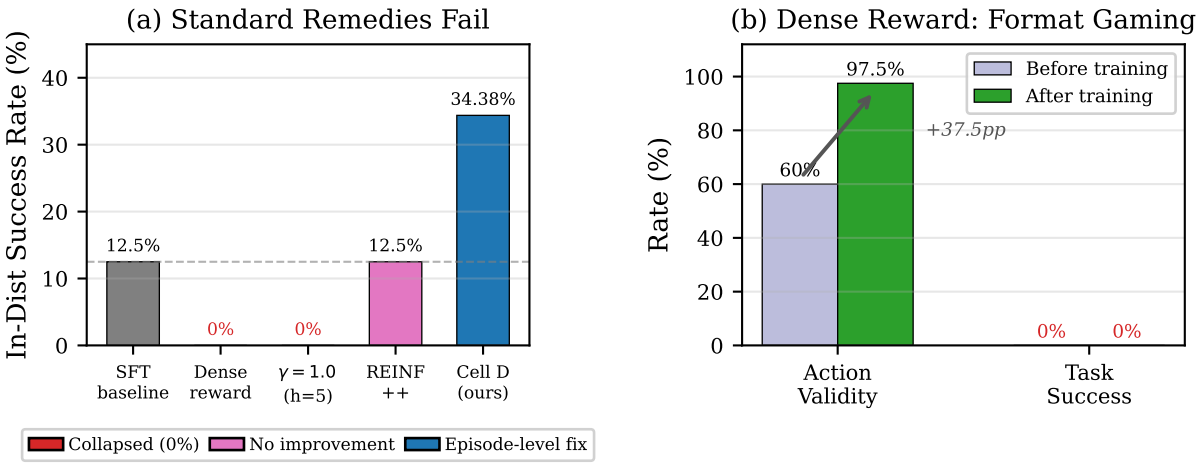


Figure 4: (a) Standard remedies for step-independent training: dense reward shaping and discount factor γ=1.0 both collapse to 0%; REINFORCE++ matches the SFT baseline without improvement. Only episode-level training (Cell D) exceeds baseline. (b) Dense reward gaming: action validity rises from 60% to 97.5% while task success remains 0%—the agent maximizes format-level rewards without solving tasks.

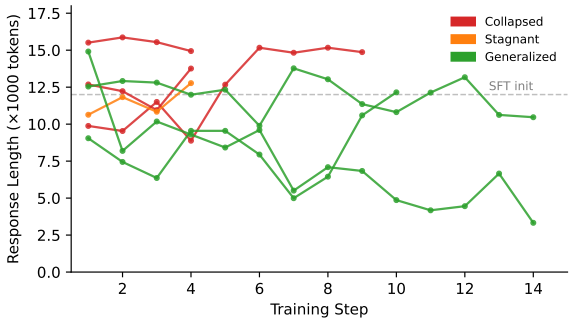


Figure 5: Response length over training for 7 seeds under episode-level GRPO. Collapsed seeds (red) exhibit length inflation to 14,000–16,000 tokens. Generalized seeds (green) maintain or reduce response length. Stagnant seed (orange, seed100) remains near the SFT initialization.